Status	Finished
Started	Saturday, 24 October 2026, 10:41 PM
Completed	Saturday, 24 October 2026, 10:45 PM
Duration	3 mins 18 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 0.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It simplifies the sorting process by using only one pass
- c. It only works on small datasets
- d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. sorted()
- d. order()

The correct answer is: sorted()

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Sorting data in a non-sequential manner
c. Bubbling up the largest element to its correct position with each pass
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Distribution
c. Insertion
d. Extraction

The correct answer is:
Insertion

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Merge sort
c. Selection sort
d. Quick sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower

b. It allows for faster searching

c. It has no effect on the search operation

d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()

b. order()

c. sort()

d. sorted()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Searching
d. Sorting

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Searching for data in a list
c. Deleting data from a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. The parts are merged without sorting
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently